Predicting the effects of behavioral interventions on humans with LLMs

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We test the potential of predicting human behavior in survey experiments with behavioral clones—large language model (LLM) generated artificial agents mimicking human participants. To evaluate this potential, we predict the outcomes of a large-scale field experiment (N = ~22,000) to be conducted in the United States. The experiment will test 20 text-based interventions designed to increase trust in climate scientists, assessing a broad range of secondary, outcomes including self-reported attitudes (e.g., support for funding of climate science) and behavioral measures (e.g., donations to a scientific organization). Before the human study is fielded, we will simulate it with behavioral clones using demographically representative U.S. participant profiles. We will lock the simulated dataset and analysis code. Once human data are available, we will compare clone-based predictions to human results for average treatment effect and subgroup heterogeneity. We will also compare how well the LLMS produce outcome distributions, and whether they overemphasize stereotype responses. Our study will provide a large-scale, credible benchmark test of LLMs' potential—and limitations—as tools for simulating human subjects in social science research. All analyses in this preregistration are demonstrated on simulated placeholder data with identical structure to the real datasets; no human outcome data have been observed at the time of registration.

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```r
# packages
library(tidyverse)
library(sandwich)
library(lmtest)
library(marginaleffects)
library(broom)
library(MetBrewer)
library(patchwork)
library(gt)
library(tinytable)
library(here)

# custom functions
source(here("R/functions/statistics.R"))
source(here("R/functions/plots.R"))
source(here("R/functions/tables.R"))
source(here("R/functions/reporting.R"))
```

```r
# simulated (random) data
human_data <- readRDS(here("data/simulation_preregistration/human_data_preregistration.rds"))
llm_data   <- readRDS(here("data/simulation_preregistration/llm_data_preregistration.rds"))

# actual data later
# human_data  <- readRDS("data/cleaned_llm.rds")
# llm_data <- readRDS("data/cleaned_llm.rds")
```

> **Important**
>
> **Placeholder data.** Both datasets used throughout are simulated random noise with the correct structure. The analysis code will be applied to real human and LLM clone data without modification once both are available.

Motivation

The aim of this project is to understand how well large language models (LLMs) can simulate human behavior in survey experiments.

Large language models (LLMs) are trained on extensive human-generated data. Consequently, they reproduce patterns of human judgment and reasoning (Binz and Schulz 2023; Aher, Arriaga, and Kalai 2023). In controlled, text-based interactions, humans sometimes struggle to distinguish between responses generated by LLMs and those produced by other humans (Jones et al. 2025; but see Pagan et al. 2025).

This apparent capability of LLMs to mimic human behavior has generated substantial interest among social scientists. Running surveys and experiments with human subjects is costly and time-consuming. If LLMs could reliably approximate human behavior, they would offer a fast and low-cost alternative. While it seems unrealistic at this point that LLMs could substitute human subjects entirely (see, e.g., Schröder et al. 2025), they might still open up substantial new opportunities for social science research. Academic publishing systems tend to incentivize positive results (Nosek, Spies, and Motyl 2012; Smaldino and McElreath 2016), which can discourage the testing of novel or high-risk ideas. Approximative LLM simulations could serve as a low-risk tool for pretesting such novel ideas before committing resources to human data collection. LLMs could also help testing the boundaries of established theories. Scientific theories—at least in the social sciences—are often highly context-dependent. To improve theory building, researchers have called for more systematic exploration of these contextual variations (Almaatouq et al. 2022). LLMs could make such exploration more feasible: they could simulate a multitude of variations of an experiment, thereby pointing researchers to critical context factors that could be tested in follow-up studies on human participants.

Recent studies have demonstrated promising results for LLMs in simulating public opinion (Kaiser et al. 2025; Wang, Morgenstern, and Dickerson 2025; Argyle et al. 2023; Durmus et

al. 2024; Santurkar et al. 2023; Bisbee et al. 2024; Shrestha et al. 2024) and in predicting the outcomes of experiments with human participants (Hewitt et al. 2025; Chen, Hu, and Lu 2025; Hu et al. 2025; Guo et al. 2025; Lippert et al. 2024; Bisbee et al. 2024). At the same time, other work has identified important limitations of LLM simulations, including systematic misrepresentation of certain identity groups (Wang, Morgenstern, and Dickerson 2025), a general tendency of overestimating experimental effects (Doudkin, Pataranutaporn, and Maes 2025; Cui, Li, and Zhou 2024), and high volatility of simulation results, depending on model and prompt choice (Schröder et al. 2025).

Currently, interpreting these mixed findings is difficult, as most existing studies face at least one of the following limitations. First, some studies rely on already published experiments, raising concerns that LLMs may have been trained on the very data they are asked to simulate. These studies might only test LLMs capacity to reproduce, but not to predict. Second, even studies that plausibly rely on unpublished data are rarely preregistered. Given that LLM performance is highly sensitive to model choice and prompting, preregistered predictions are crucial for credible evaluation. Third, some studies operate at the experiment level, providing LLMs with a full description of the study and requesting a single aggregate prediction. While informative about average effects, this approach precludes the analysis of individual-level heterogeneity.

In this project, we aim to provide a robust test of how well LLMs can simulate human behavior in survey experiments. We address limitations of previous studies by leveraging a unique opportunity to evaluate LLM simulations in the context of a large-scale field experiment. I am currently leading a study involving approximately 21,000 participants from the United States (US), testing 20 text-based interventions designed to increase trust in climate scientists. The study assesses a broad range of outcomes, including self-reported attitudes (e.g., trust in scientists) and behavioral measures (e.g., donations to a scientific organization). Before running the study with human participants, we will use multiple LLMs and prompting strategies to simulate individual-level survey responses representative of the US population. We will then collect the human data and test how closely they match the data simulated by the LLMs. This design allows for a large-scale, credible benchmark test of LLMs' potential—and limitations—as tools for simulating human subjects in social science research.

Research Questions

The main outcome of the megastudy is trust in climate scientists, but several secondary and tertiary outcomes are preregistered. Analyses are organized in three sections that mirror one another: we first examine average effects, then subgroup differences, and finally demographic calibration at the individual level.

Section 1 — Average Treatment Effects. We ask (a) how well LLM clones predict the average treatment effect (ATE) of each intervention across all outcomes, and (b) whether clones reproduce the full response distribution — not just the mean — in the control condition, across all continuous outcomes.

Section 2 — Subgroup Effects. We ask (a) whether clones recover differences in intervention effects across six demographic moderators (gender, age, race, education, income, partisan identity) for all outcomes, and (b) whether within-subgroup response distributions in the control condition match between humans and clones for the primary outcome.

Section 3 — Demographic Stereotyping. We ask whether individual clone responses are calibrated to the assigned demographic profile: (a) do clones reproduce the correct baseline response level for each demographic group, and (b) do demographic variables predict clone responses with the same magnitude as in humans?

Data

Human data

LLM generated data

Clone profiles were constructed by drawing 9,000 individuals from four recent waves of the General Social Survey (GSS; 2018, 2021, 2022, 2024). Within-year normalized sampling weights were applied so that each wave contributes equal sampling probability to the pool, while preserving within-year relative weights. Each profile was assigned to exactly one experimental condition (between-subjects, matching the human study design). Profiles encode the following demographic characteristics: age, sex, race/ethnicity (racial self-identification and Hispanic origin), educational attainment, household income, household size, self-identified social class, U.S. region, partisan identification, political ideology, religious affiliation, religious intensity, and scientific consensus acceptance. Each clone then completed the survey experiment under the same conditions as human participants.

Two limitations of the current LLM dataset should be noted. First, the racial coding uses the GSS `race` variable, which distinguishes only White, Black, and Other; Asian respondents are subsumed under "Other." A revised profile generation script uses the more detailed Census-style `racecen1` variable, which enables a distinct Asian category, but the LLM responses in this study were generated from profiles using the original three-category variable. Second, the current profiles omit the GSS `srcbelt` variable, which encodes community type (urban, suburban, rural). Both limitations affect only the demographic detail of clone profiles, not the experimental design or outcome measurement.

```r
profiles <- read_csv(here("clone_profiles", "profiles.csv"), show_col_types = FALSE)

meta_vars <- c("profile_id", "year", "id", "wtssps", "w_pooled",
               "profile_basic", "profile_detailed")
prof_vars <- setdiff(names(profiles), meta_vars)
yrs       <- sort(unique(profiles$year))
```

```r
yr_ns <- profiles |> count(year) |> deframe()

comp_tbl <- prof_vars |>
  map_dfr(\(v) {
    row <- tibble(Variable = v)
    for (yr in yrs) {
      d <- profiles |> filter(year == yr) |> pull(all_of(v))
      row[[paste0(yr, " (N=", yr_ns[[as.character(yr)]], ")")]] <- sum(is.na(d))
    }
    row
  })

comp_tbl |>
  tt() |>
  style_tt(j = 1, align = "l") |>
  style_tt(j = seq(2, ncol(comp_tbl)), align = "r") |>
  style_tt(i = 0, bold = TRUE)
```

Table 1: Missing values per variable by GSS wave. Column headers show the wave year and total N for that wave in brackets.

Variable	2018 (N=1911)	2021 (N=2667)	2022 (N=2176)	2024 (N=2246)
age	0	0	0	0
sex	0	0	0	0
racecen1	0	0	0	0
hispanic	0	0	0	0
degree	0	0	0	0
income16	0	0	0	0
hompop	0	61	1093	1126
class	0	0	0	0
region	0	0	0	0
srcbelt	0	0	0	0
partyid	0	0	0	0
polviews	0	0	0	0
relig	0	0	0	0
reborn	24	905	29	40
reliten	6	0	1033	1288
consci	654	902	764	796

```r
race_var <- if ("racecen1" %in% names(profiles)) "racecen1" else "race"

age_tbl <- profiles |>
  mutate(Category = cut(age,
    breaks = c(17, 29, 44, 59, Inf),
    labels = c("18-29", "30-44", "45-59", "60+")
  )) |>
  count(Category) |>
  mutate(pct = round(100 * n / sum(n), 1), group = "Age")

sex_tbl <- profiles |>
  count(Category = sex) |>
  mutate(pct = round(100 * n / sum(n), 1), group = "Sex")

race_tbl <- profiles |>
  count(Category = .data[[race_var]]) |>
```

```r
  mutate(pct = round(100 * n / sum(n), 1), group = "Race / Ethnicity")

demo_tbl <- bind_rows(age_tbl, sex_tbl, race_tbl) |>
  mutate(Category = as.character(Category))

demo_tbl |>
  select(Category, n, pct) |>
  rename(N = n, "%" = pct) |>
  tt(width = 0.5) |>
  group_tt(i = list(
    "Age"              = which(demo_tbl$group == "Age")[1],
    "Sex"              = which(demo_tbl$group == "Sex")[1],
    "Race / Ethnicity" = which(demo_tbl$group == "Race / Ethnicity")[1]
  )) |>
  style_tt(align = "l") |>
  style_tt(i = 0, bold = TRUE) |>
  style_tt(i = "groupi", bold = TRUE) |>
  format_tt(escape = TRUE)
```

Table 2: Distribution of key quota variables in the LLM clone profile pool (N = 9,000).

Category	N	%
Age		
18–29	1271	14.1
30–44	2481	27.6
45–59	2232	24.8
60+	3016	33.5
Sex		
female	4748	52.8
male	4252	47.2
Race / Ethnicity		
american indian or alaska native	112	1.2
asian	374	4.2
black or african american	1254	13.9
hispanic	411	4.6
other pacific islander	20	0.2
some other race	73	0.8
white	6756	75.1

Measures

```
outcomes_primary <- "trust_multidimensional"

outcomes_secondary <- c(
  "trust_post", "distrust_post",
  #"donation_ams",
  "funding_perceptions", "policy_role_mean", "inst_trust_mean"
)
```

```r
outcomes_binary <- "newsletter_signup"

outcomes_tertiary <- c(
  "belief_post", "concern_mean", "policy_general",
  "policy_specific_mean", "behavior_mean"
)

outcomes_continuous <- c(outcomes_primary, outcomes_secondary, outcomes_tertiary)

# Illustrative subset for by-outcome breakdowns (distributions, moderator detail)
outcomes_illustrative <- c(
  "trust_multidimensional",
  "donation_ams",
  "funding_perceptions",
  "policy_general"
)

trust_dimensions <- c(
  "trust_competence", "trust_integrity", "trust_benevolence", "trust_openness"
)

trust_items <- c(
  paste0("trust_competence_",  1:3),
  paste0("trust_integrity_",   1:3),
  paste0("trust_benevolence_", 1:3),
  paste0("trust_openness_",    1:3)
)

moderators_cat <- c("gender", "age_band", "race", "education", "income", "party")

outcome_label_map <- c(
  trust_multidimensional = "Multidimensional trust (primary)",
  trust_post             = "Overall trust",
  distrust_post          = "Distrust",
  donation_ams           = "Donation to AMS ($)",
  newsletter_signup      = "Newsletter sign-up",
  funding_perceptions    = "Funding perceptions",
  policy_role_mean       = "Policy role",
  inst_trust_mean        = "Institutional trust",
  belief_post            = "Climate belief",
  concern_mean           = "Climate concern",
  policy_general         = "General climate policy",
```

```
  policy_specific_mean   = "Specific climate policy",
  behavior_mean          = "Pro-climate behavior"
)
```

> **Note**
>
> **Illustrative outcomes.** Analyses that break down results by outcome — response distributions (Section 1) and subgroup heterogeneity (Section 2) — report results for an illustrative subset of four outcomes spanning the primary, secondary, and tertiary tiers: multidimensional trust (primary), donation to AMS, funding perceptions, and general climate policy. The preregistered ATE comparison (Section 1, ATE Recovery) covers all outcomes.

Comparison Functions

Two helper functions operationalize the comparison metrics. They are applied symmetrically to human and LLM clone model outputs throughout the document.

Comparison metrics

`compare_estimates()` is used throughout — for ATEs (RQ1), moderator interaction estimates (RQ3), and secondary analyses. It takes two tidy model result tibbles and a `join_by` argument specifying the key columns, and computes six metrics:

- **Spearman** — rank correlation of point estimates. Answers: do clones rank the 20 interventions in the same order as humans? A value of 1 means perfect rank agreement; 0 means no correspondence.
- **Pearson r** — linear correlation of point estimates. Sensitive to outliers but captures linear proportionality.
- **RMSE** — root mean squared error. Captures average magnitude error in the same units as the outcome.
- **Directional agreement** — % of estimates with the same sign. Answers: do clones at least get the direction of each effect right?
- **Inferential agreement** — % of estimates with the same BH-adjusted conclusion (significantly positive, significantly negative, or non-significant at $= .05$). Answers: would a researcher using clones reach the same inferential conclusion as one using human data?
- **TOST (%)** — % of individual estimate pairs that pass an equivalence test with preregistered bound $\Delta_k = 0.5 \times |\text{estimate_human_k}|$. Equivalence is declared (two one-sided tests, $= .05$) when the clone estimate falls within ±50% of the human estimate. Note: when human estimates are near zero, Δ_k shrinks accordingly, making equivalence harder to achieve.

```r
print_a_function_from_file("compare_estimates")
```

```r
compare_estimates <- function(h_result, l_result,
                              join_by = "condition",
                              tost_delta_pct = 0.5) {

  # Standardize columns; SE derived from std.error or from 95% CI width if absent
  prep <- function(df, suffix) {
    df |>
      select(all_of(join_by), estimate) |>
      mutate(
        se    = if ("std.error" %in% names(df)) df$std.error
                else if (all(c("conf.low", "conf.high") %in% names(df)))
                  (df$conf.high - df$conf.low) / (2 * 1.96)
                else NA_real_,
        p_adj = if ("p.value_adjusted" %in% names(df)) df$p.value_adjusted
                else NA_real_
      ) |>
      rename_with(~ paste0(.x, "_", suffix), c(estimate, se, p_adj))
  }

  joined <- inner_join(prep(h_result, "h"), prep(l_result, "l"), by = join_by)

  # Three-level inferential category based on BH-adjusted p and direction
  infer_cat <- function(est, p_adj) {
    case_when(
      p_adj < 0.05 & est > 0 ~ "sig_positive",
      p_adj < 0.05 & est < 0 ~ "sig_negative",
      TRUE                   ~ "not_significant"
    )
  }

  # TOST: equivalence declared when |ATE_h - ATE_l| < delta = tost_delta_pct × |ATE_h|
  # p_tost is the maximum of the two one-sided p-values; equivalent if p_tost < 0.05
  joined |>
    mutate(
      delta      = tost_delta_pct * abs(estimate_h),
      diff       = estimate_h - estimate_l,
      se_diff    = sqrt(se_h^2 + se_l^2),
      p_lower    = pnorm((diff + delta) / se_diff, lower.tail = FALSE),
      p_upper    = pnorm((diff - delta) / se_diff),
      p_tost     = pmax(p_lower, p_upper),
```

```
      equivalent = p_tost < 0.05,
      same_infer = infer_cat(estimate_h, p_adj_h) == infer_cat(estimate_l, p_adj_l)
    ) |>
    summarise(
      spearman_rho    = cor(estimate_h, estimate_l, method = "spearman",
                            use = "pairwise.complete.obs"),
      pearson_r       = cor(estimate_h, estimate_l, use = "pairwise.complete.obs"),
      rmse            = sqrt(mean((estimate_h - estimate_l)^2, na.rm = TRUE)),
      directional_pct = mean(sign(estimate_h) == sign(estimate_l), na.rm = TRUE) * 100,
      inferential_pct = if (!anyNA(p_adj_h) && !anyNA(p_adj_l))
        mean(same_infer, na.rm = TRUE) * 100 else NA_real_,
      tost_pct        = mean(equivalent, na.rm = TRUE) * 100
    )
}
```

Distribution comparison metrics

`compare_distributions()` compares response distributions for a single condition and outcome using three metrics:

- **OVL (overlapping coefficient)** — proportion of area where the two kernel density estimates overlap; 0 = no overlap, 1 = identical distributions; primary metric for RQ2
- **KS D-statistic** — maximum absolute difference between the two empirical CDFs; 0 = identical
- **Variance ratio** — LLM clone variance divided by human variance; 1 = equal spread

```
print_a_function_from_file("compute_ovl")
```

```
compute_ovl <- function(x, y, n_grid = 512) {
  lo <- min(c(x, y)); hi <- max(c(x, y))
  if (lo == hi) return(NA_real_)
  # Evaluate both KDEs on the same grid, then integrate the overlap area
  d_h <- density(x, from = lo, to = hi, n = n_grid)
  d_l <- density(y, from = lo, to = hi, n = n_grid)
  sum(pmin(d_h$y, d_l$y)) * (hi - lo) / n_grid
}
```

Calibration regression

Correlation between human and clone ATEs tells us only whether the two effect-size vectors *covary* — whether interventions that produce large effects in humans also produce large effects

in clones. It is invariant to additive shifts and to multiplicative rescaling: clones could inflate every effect by a factor of two, or depress every effect by a constant, and Pearson (or Spearman) correlation would still be 1. For a megastudy, rank agreement is necessary but not sufficient. If clone ATEs are to be used to anchor power calculations, prioritise interventions for follow-up, or substitute for human pilot estimates, they must lie on the same *scale* as human ATEs, not merely correlate with them. Calibration regression decomposes scale agreement into two failure modes — additive bias and proportional miscalibration — that correlation by construction cannot detect.

`run_calibration()` regresses human ATEs on LLM clone ATEs: `ATE_human =  +  × ATE_clone`. The intercept captures additive bias: > 0 means clones systematically underestimate human ATEs (an offset must be added to recover them), < 0 means they overestimate. The slope captures proportionality: $= 1$ is perfect scaling (clone effects move 1:1 with human effects), > 1 means clone effects are compressed relative to humans (too small in absolute magnitude), < 1 means they are exaggerated. We report 95% confidence intervals for both parameters and R^2 as the share of human-ATE variance linearly recoverable from clone ATEs.

We then test calibration at two levels of strictness. The **joint F-test of (= 0, = 1)** is the omnibus test of perfect calibration: it asks whether clone ATEs can be read directly as estimates of human ATEs, with no correction at all. The **slope-only F-test of (= 1)** asks the weaker but, for our purposes, more diagnostic question — whether clone effects are *proportionally* calibrated, i.e. whether a one-unit difference in a clone ATE corresponds to a one-unit difference in a human ATE, while permitting a constant additive offset . The slope-only test is meaningful in addition to the joint test for three reasons. First, most downstream uses of clone data are invariant to additive bias: between-intervention contrasts, rank orderings, top-K decisions, and effect-spread-based power calculations all difference out a constant , so $= 1$ is what those uses actually require. Second, an additive bias is the easiest miscalibration to correct post-hoc — a single human anchor condition is enough — whereas a slope distortion reflects a deeper failure in how clones translate to humans and cannot be removed without re-estimating itself. Third, when the joint test rejects, comparing the joint p-value to the slope-only p-value localises the failure: a rejected joint test together with a non-rejected $= 1$ implies the problem is offset (relatively benign and correctable), whereas a rejected $= 1$ implies proportional miscalibration and a more substantive limit on what clone ATEs can be used for.

```r
print_a_function_from_file("run_calibration")
```

```r
run_calibration <- function(h_result, l_result) {
  joined <- inner_join(
    h_result |> select(condition, estimate_h = estimate),
    l_result |> select(condition, estimate_l = estimate),
    by = "condition"
```

```r
  )

  # ATE_h =   +   × ATE_l
  fit   <- lm(estimate_h ~ estimate_l, data = joined)
  coefs <- broom::tidy(fit, conf.int = TRUE)

  tibble(
    alpha     = coefs$estimate[1],
    alpha_lo  = coefs$conf.low[1],
    alpha_hi  = coefs$conf.high[1],
    beta      = coefs$estimate[2],
    beta_lo   = coefs$conf.low[2],
    beta_hi   = coefs$conf.high[2],
    r_squared = broom::glance(fit)$r.squared,
    # Joint F-test  H0:   = 0 AND   = 1  (perfect calibration on the original scale)
    p_joint   = car::linearHypothesis(
                  fit, c("(Intercept) = 0", "estimate_l = 1")
                )$`Pr(>F)`[2],
    # Slope-only F-test  H0:   = 1  (proportional calibration; constant offset allowed)
    p_beta_1  = car::linearHypothesis(fit, "estimate_l = 1")$`Pr(>F)`[2]
  )
}
```

1. Average Treatment Effects

ATE Recovery

For each intervention, the average treatment effect (ATE) is the difference in mean outcome between participants assigned to that intervention and those in the control condition. We estimate ATEs by running `run_main_treatment_model()` — OLS with HC2 heteroskedasticity-robust standard errors and BH-adjusted p-values — on human and LLM clone data separately for each preregistered outcome. For `newsletter_signup` (binary), we use `run_main_treatment_model_binary()`, which returns marginal effects on the probability scale.

Good prediction means the clones rank the interventions in the same order as humans (high Spearman), and that individual ATE pairs fall within a preregistered tolerance band (high TOST %). We first report metrics pooled across all outcomes — 240 estimate pairs in total — to maximize statistical power, then break down by outcome and zoom in on the primary outcome. All models are fit first.

```r
# Run and store all models (avoids re-fitting in later chunks)
ate_model_results <- map(outcomes_continuous, function(out) {
  list(
    h = run_main_treatment_model(human_data, outcome = out),
    l = run_main_treatment_model(llm_data,   outcome = out)
  )
}) |> set_names(outcomes_continuous)

# Binary outcome: marginal effects on probability scale
h_bin <- run_main_treatment_model_binary(human_data, "newsletter_signup")
l_bin <- run_main_treatment_model_binary(llm_data,   "newsletter_signup")

# References to primary outcome models for the primary subsection below
h_primary <- ate_model_results[["trust_multidimensional"]]$h
l_primary <- ate_model_results[["trust_multidimensional"]]$l

# Extract ATE pairs with SE and adjusted p-value from a model result tibble
prep_ate <- function(df) {
  df |>
    select(condition, estimate) |>
    mutate(
      se    = if ("std.error"        %in% names(df)) df$std.error        else NA_real_,
      p_adj = if ("p.value_adjusted" %in% names(df)) df$p.value_adjusted else NA_real_
    )
}

# Pool ATE pairs across all outcomes (continuous + binary)
pooled_ates <- bind_rows(
  map_dfr(outcomes_continuous, function(out) {
    inner_join(
      prep_ate(ate_model_results[[out]]$h) |>
        rename(estimate_h = estimate, se_h = se, p_adj_h = p_adj),
      prep_ate(ate_model_results[[out]]$l) |>
        rename(estimate_l = estimate, se_l = se, p_adj_l = p_adj),
      by = "condition"
    ) |> mutate(outcome = out)
  }),
  inner_join(
    prep_ate(h_bin$marginal_effects) |>
      rename(estimate_h = estimate, se_h = se, p_adj_h = p_adj),
    prep_ate(l_bin$marginal_effects) |>
      rename(estimate_l = estimate, se_l = se, p_adj_l = p_adj),
```

```r
    by = "condition"
  ) |> mutate(outcome = "newsletter_signup")
) |>
  mutate(
    tier = case_when(
      outcome == outcomes_primary                          ~ "Primary",
      outcome %in% c(outcomes_secondary, outcomes_binary) ~ "Secondary",
      TRUE                                                 ~ "Tertiary"
    ),
    outcome_label = outcome_label_map[outcome]
  )

# Per-outcome summary metrics
ate_metrics <- map_dfr(outcomes_continuous, function(out) {
  compare_estimates(ate_model_results[[out]]$h, ate_model_results[[out]]$l) |>
    mutate(outcome = out)
})

calib_metrics <- map_dfr(outcomes_continuous, function(out) {
  run_calibration(ate_model_results[[out]]$h, ate_model_results[[out]]$l) |>
    mutate(outcome = out)
})

ate_metrics <- bind_rows(
  ate_metrics,
  compare_estimates(h_bin$marginal_effects, l_bin$marginal_effects) |>
    mutate(outcome = "newsletter_signup")
) |>
  mutate(
    tier = case_when(
      outcome == outcomes_primary                          ~ "Primary",
      outcome %in% c(outcomes_secondary, outcomes_binary) ~ "Secondary",
      TRUE                                                 ~ "Tertiary"
    ),
    outcome_label = outcome_label_map[outcome]
  )

calib_metrics <- bind_rows(
  calib_metrics,
  run_calibration(h_bin$marginal_effects, l_bin$marginal_effects) |>
    mutate(outcome = "newsletter_signup")
) |>
```

```r
  mutate(
    tier = case_when(
      outcome == outcomes_primary                          ~ "Primary",
      outcome %in% c(outcomes_secondary, outcomes_binary) ~ "Secondary",
      TRUE                                                 ~ "Tertiary"
    ),
    outcome_label = outcome_label_map[outcome]
  )

# Pooled summary row - prepended to ate_metrics so it appears at the top of tbl-rq1
pooled_row <- pooled_ates |>
  mutate(
    delta      = 0.5 * abs(estimate_h),
    diff       = estimate_h - estimate_l,
    se_diff    = sqrt(se_h^2 + se_l^2),
    p_lower    = pnorm((diff + delta) / se_diff, lower.tail = FALSE),
    p_upper    = pnorm((diff - delta) / se_diff),
    p_tost     = pmax(p_lower, p_upper),
    equivalent = p_tost < 0.05,
    same_sign  = sign(estimate_h) == sign(estimate_l),
    infer_h    = case_when(
      !is.na(p_adj_h) & p_adj_h < 0.05 & estimate_h > 0 ~ "sig_positive",
      !is.na(p_adj_h) & p_adj_h < 0.05 & estimate_h < 0 ~ "sig_negative",
      TRUE ~ "not_significant"
    ),
    infer_l    = case_when(
      !is.na(p_adj_l) & p_adj_l < 0.05 & estimate_l > 0 ~ "sig_positive",
      !is.na(p_adj_l) & p_adj_l < 0.05 & estimate_l < 0 ~ "sig_negative",
      TRUE ~ "not_significant"
    )
  ) |>
  summarise(
    spearman_rho    = cor(estimate_h, estimate_l, method = "spearman",
                          use = "pairwise.complete.obs"),
    pearson_r       = cor(estimate_h, estimate_l, use = "pairwise.complete.obs"),
    rmse            = NA_real_,
    directional_pct = mean(same_sign,  na.rm = TRUE) * 100,
    inferential_pct = mean(infer_h == infer_l, na.rm = TRUE) * 100,
    tost_pct        = mean(equivalent, na.rm = TRUE) * 100
  ) |>
  mutate(outcome = "pooled", tier = "All outcomes", outcome_label = "All outcomes")
```

```r
ate_metrics <- bind_rows(pooled_row, ate_metrics)
```

All outcomes

The scatter below shows all 240 estimate pairs — one per intervention × outcome combination — colored by outcome tier. Points above the diagonal indicate clone overestimation; points below indicate underestimation. Because estimates from different outcomes are on different scales, the spread also reflects scale variation across outcomes.

The comparison table below includes a pooled summary row at the top, followed by per-outcome rows grouped by tier. RMSE is left blank for the pooled row because it is not comparable across outcomes on different scales; all other metrics are scale-invariant. The calibration regression (second table) and faceted scatter (third figure) are per-outcome only.

```r
tier_colors <- c("Primary" = "#C0392B", "Secondary" = "#2980B9", "Tertiary" = "#7F8C8D")

pooled_ates |>
  ggplot(aes(x = estimate_h, y = estimate_l, color = tier)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed",
              color = "grey60", linewidth = 0.5) +
  geom_hline(yintercept = 0, linetype = "dotted", color = "grey80") +
  geom_vline(xintercept = 0, linetype = "dotted", color = "grey80") +
  geom_point(size = 2, alpha = 0.65) +
  scale_color_manual(values = tier_colors, name = "Outcome tier") +
  labs(x = "Human ATE", y = "LLM clone ATE") +
  plot_theme +
  theme(legend.position = "top")
```

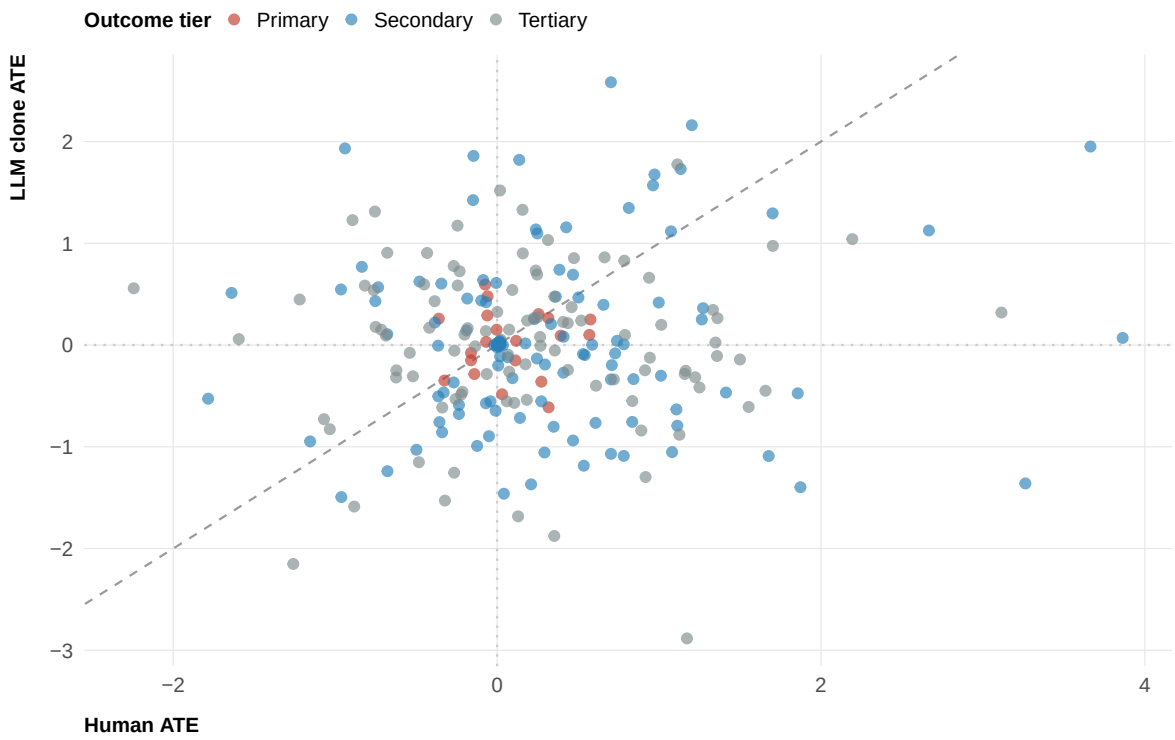

Figure 1: Human vs. LLM clone ATEs pooled across all outcomes. Each point is one intervention × outcome pair (N = 240). The dashed line is perfect agreement; dotted lines mark zero. Color indicates outcome tier.

```
ate_metrics |>
  arrange(match(tier, c("All outcomes", "Primary", "Secondary", "Tertiary")), outcome_label)
  select(Tier = tier, Outcome = outcome_label,
         `Spearman  ` = spearman_rho, `Pearson r` = pearson_r, RMSE = rmse,
         `Directional (%)` = directional_pct, `Inferential (%)` = inferential_pct,
         `TOST (%)` = tost_pct) |>
  gt(groupname_col = "Tier") |>
  fmt_number(columns = c(`Spearman  `, `Pearson r`, RMSE), decimals = 3) |>
  sub_missing(columns = "RMSE", missing_text = "-") |>
  fmt_number(columns = c(`Directional (%)`, `Inferential (%)`, `TOST (%)`), decimals = 1) |>
  tab_style(
    style     = cell_text(weight = "bold"),
    locations = cells_row_groups()
  )
```

Table 3: RQ1 comparison metrics. The 'All outcomes' row pools all estimate pairs across outcomes; RMSE (—) is omitted there because outcomes differ in scale. Per-outcome rows each have N = 20 intervention pairs. TOST: % of pairs achieving equivalence (Δ = 50% of human ATE).

Outcome	Spearman	Pearson r	RMSE	Directional (%)	Inferential (%)
All outcomes					
All outcomes	-0.003	0.056	—	50.4	94.6
Primary					
Multidimensional trust (primary)	0.023	0.032	0.408	50.0	100.0
Secondary					
Distrust	-0.077	-0.161	1.450	70.0	90.0
Funding perceptions	0.017	-0.009	1.618	50.0	95.0
Institutional trust	-0.205	-0.140	1.224	30.0	100.0
Newsletter sign-up	0.266	0.223	0.016	65.0	100.0
Overall trust	0.096	0.135	1.257	60.0	100.0
Policy role	-0.122	-0.127	0.700	30.0	100.0
Tertiary					
Climate belief	-0.117	-0.035	1.395	45.0	100.0
Climate concern	0.021	0.099	1.140	40.0	100.0
General climate policy	-0.005	0.026	1.345	60.0	100.0
Pro-climate behavior	-0.409	-0.477	1.234	40.0	50.0
Specific climate policy	0.301	0.279	0.377	65.0	100.0

```
calib_metrics |>
  arrange(match(tier, c("Primary", "Secondary", "Tertiary")), outcome_label) |>
  mutate(
    beta_ci  = paste0("[", round(beta_lo,  2), ", ", round(beta_hi,  2), "]"),
    alpha_ci = paste0("[", round(alpha_lo, 2), ", ", round(alpha_hi, 2), "]")
  ) |>
  select(Tier = tier, Outcome = outcome_label,
         alpha, alpha_ci, beta, beta_ci, r_squared, p_beta_1) |>
  gt(groupname_col = "Tier") |>
  cols_label(alpha = " ", alpha_ci = "95% CI ( )",
             beta = " ", beta_ci = "95% CI ( )",
             r_squared = "R²", p_beta_1 = "p ( =1)") |>
  fmt_number(columns = c(alpha, beta, r_squared), decimals = 3) |>
```

```
  fmt_number(columns = p_beta_1, decimals = 3) |>
  tab_style(
    style     = cell_text(weight = "bold"),
    locations = cells_row_groups()
  )
```

Table 4: Calibration regression by outcome. : intercept (systematic bias, with 95% CI). : slope (proportionality; 1 = perfect, with 95% CI). R^2: variance explained. p(=1): F-test of perfect calibration.

Outcome		95% CI ( )		95% CI ( )	R^2	p (=1)
Primary						
Multidimensional trust (primary)	0.077	[-0.05, 0.21]	0.027	[-0.39, 0.44]	0.001	0.000
Secondary						
Distrust	1.043	[0.31, 1.78]	-0.194	[-0.78, 0.39]	0.026	0.000
Funding perceptions	0.455	[-0.18, 1.09]	-0.014	[-0.79, 0.76]	0.000	0.013
Institutional trust	0.184	[-0.27, 0.64]	-0.154	[-0.7, 0.39]	0.020	0.000
Newsletter sign-up	0.009	[0, 0.02]	0.170	[-0.2, 0.54]	0.050	0.000
Overall trust	0.272	[-0.23, 0.77]	0.140	[-0.37, 0.65]	0.018	0.002
Policy role	-0.074	[-0.27, 0.12]	-0.093	[-0.45, 0.27]	0.016	0.000
Tertiary						
Climate belief	0.557	[-0.02, 1.14]	-0.048	[-0.72, 0.63]	0.001	0.004
Climate concern	-0.407	[-0.79, -0.03]	0.101	[-0.4, 0.6]	0.010	0.001
General climate policy	-0.225	[-0.71, 0.26]	0.020	[-0.37, 0.41]	0.001	0.000
Pro-climate behavior	0.897	[0.7, 1.09]	-0.672	[-1.29, -0.06]	0.228	0.000
Specific climate policy	0.039	[-0.1, 0.17]	0.238	[-0.17, 0.64]	0.078	0.001

```
map_dfr(outcomes_continuous, function(out) {
  inner_join(
    ate_model_results[[out]]$h |> select(condition, estimate_h = estimate),
    ate_model_results[[out]]$l |> select(condition, estimate_l = estimate),
    by = "condition"
  ) |> mutate(outcome = outcome_label_map[out])
}) |>
  ggplot(aes(x = estimate_h, y = estimate_l)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed",
              color = "grey60", linewidth = 0.4, alpha = 0.35) +
```

```r
  geom_hline(yintercept = 0, linetype = "dotted", color = "grey80") +
  geom_vline(xintercept = 0, linetype = "dotted", color = "grey80") +
  geom_smooth(method = "lm", se = FALSE, color = "#2980B9", linewidth = 0.6) +
  geom_point(size = 1.5, alpha = 0.7, color = "grey20") +
  facet_wrap(~ outcome, scales = "free", ncol = 3) +
  labs(x = "Human ATE", y = "LLM clone ATE") +
  plot_theme
```

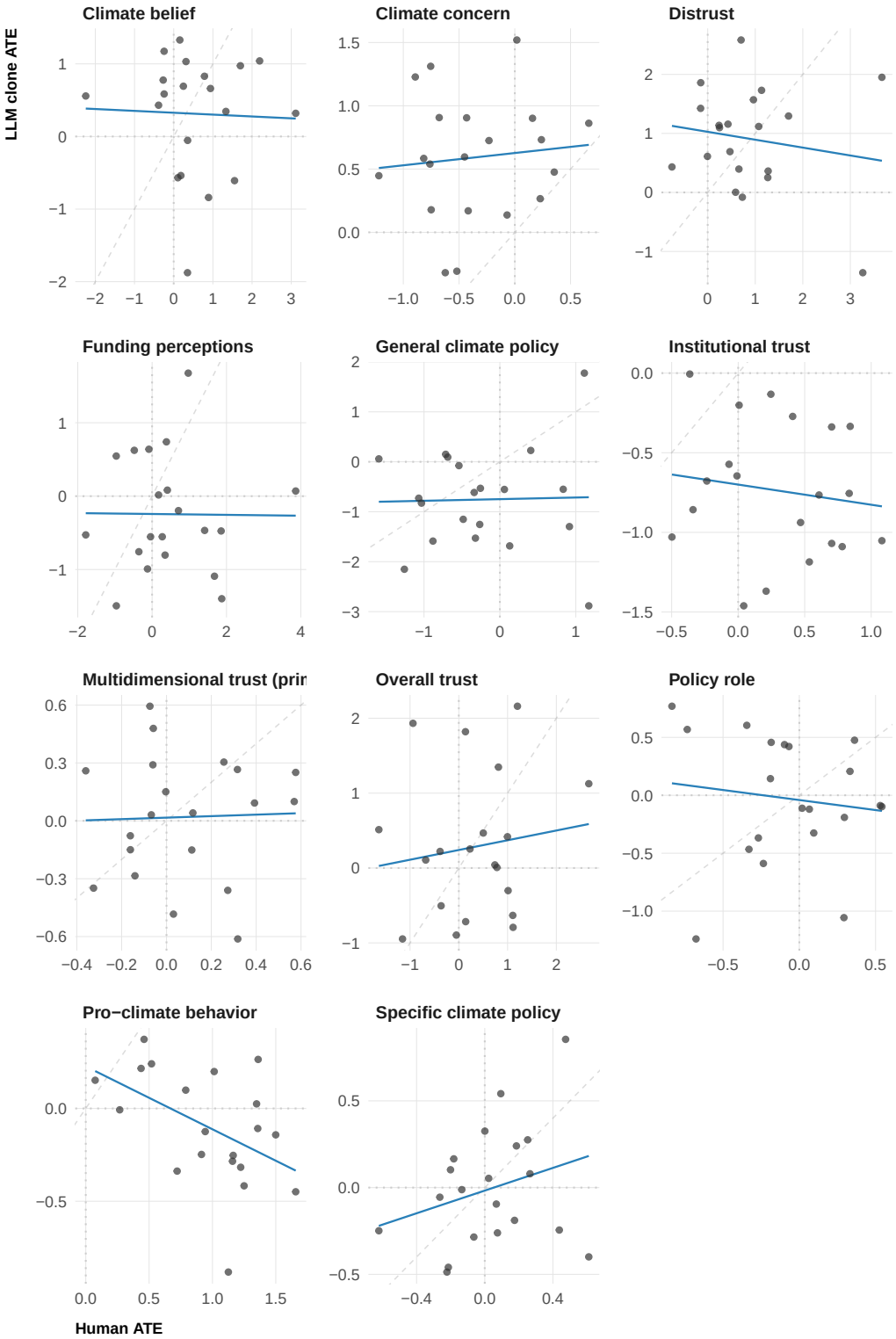

Figure 2: Human vs. LLM clone ATEs for all continuous outcomes. Each point is one of the 20 interventions; axes are free-scaled across panels. Blue line: OLS regression through the panel's points. Faint dashed line: perfect calibration (slope = 1).

Primary outcome (focused)

The primary outcome, multidimensional trust in climate scientists, is the pre-specified main endpoint — a composite averaging four subdimensions (competence, integrity, benevolence, openness). We show its ATE scatter separately to allow easy reading of individual intervention labels. The calibration regression fits `ATE_human =  +  × ATE_clone`: = 0 means no systematic bias; = 1 means clone effects are proportionally accurate. We test = 1 via a formal F-test.

```
inner_join(
  h_primary |> select(condition, estimate_h = estimate),
  l_primary |> select(condition, estimate_l = estimate),
  by = "condition"
) |>
  ggplot(aes(x = estimate_h, y = estimate_l)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed",
              color = "grey60", linewidth = 0.5, alpha = 0.35) +
  geom_hline(yintercept = 0, linetype = "dotted", color = "grey80") +
  geom_vline(xintercept = 0, linetype = "dotted", color = "grey80") +
  geom_smooth(method = "lm", se = FALSE, color = "#2980B9", linewidth = 0.7) +
  geom_point(size = 2.5, alpha = 0.8, color = "grey20") +
  geom_text(aes(label = str_remove(condition, "intervention_")),
            size = 2.5, vjust = -0.8, color = "grey40") +
  labs(x = "Human ATE (OLS estimate)", y = "LLM clone ATE (OLS estimate)") +
  plot_theme
```

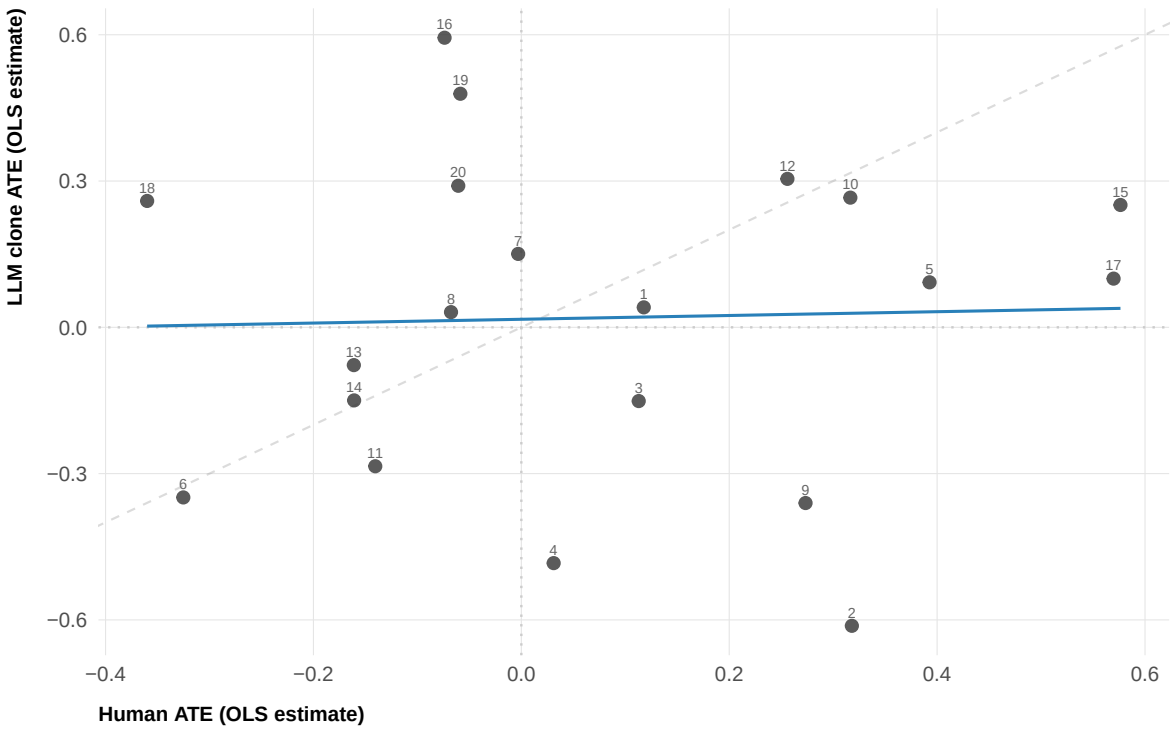

Figure 3: Human vs. LLM clone ATEs for the primary outcome (multidimensional trust). Each point is one of the 20 interventions. Blue line: OLS regression through the points. Faint dashed line: perfect calibration (slope = 1). Dotted lines mark zero.

```
run_calibration(h_primary, l_primary)
```

```
# A tibble: 1 x 9
   alpha alpha_lo alpha_hi   beta beta_lo beta_hi r_squared  p_joint p_beta_1
   <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>     <dbl>    <dbl>    <dbl>
1 0.0771  -0.0533    0.208 0.0270  -0.386   0.440   0.00105 0.000363 0.000103
```

Response Distributions

Even if clones match the average treatment effect for each intervention, they may diverge from humans in how responses are distributed: two groups with the same mean can differ in spread, skewness, or tail shape. We ask whether clones reproduce the full response distribution in the control condition — not just the mean — across all continuous outcomes. (`newsletter_signup` is excluded because kernel density estimates are not meaningful for a binary outcome.)

We compare distributions using three metrics. The **overlapping coefficient (OVL)** is the proportion of area shared between the two kernel density estimates: it equals 1 when distributions are identical and 0 when they share no support. Intuitively, imagine overlaying the two smoothed histograms — OVL is the fraction of the total area covered by both. The **Kolmogorov–Smirnov D-statistic** is the largest vertical gap between the two empirical cumulative distribution functions at any response value: 0 means the CDFs are indistinguishable everywhere, and larger values indicate a point on the scale where the distributions diverge substantially. The **variance ratio** (clone variance / human variance) captures relative spread: a ratio of 1 is ideal, > 1 means clones are more dispersed than humans, and < 1 means clones give more uniform responses.

```
control_val <- levels(human_data$condition)[1]

dist_control <- map_dfr(outcomes_illustrative, function(out) {
  compare_distributions(human_data, llm_data, out, control_val) |>
    mutate(outcome = out)
}) |>
  mutate(
    tier = case_when(
      outcome == outcomes_primary     ~ "Primary",
      outcome %in% outcomes_secondary ~ "Secondary",
      TRUE                            ~ "Tertiary"
    ),
    outcome_label = outcome_label_map[outcome]
  )
```

```
source_colors <- c("Human" = "grey30", "LLM clone" = met.brewer("Juarez", n = 10)[1])

map_dfr(outcomes_illustrative, function(out) {
  bind_rows(
    human_data |> filter(condition == control_val) |>
      select(value = all_of(out)) |>
      mutate(source = "Human",     outcome = outcome_label_map[out]),
    llm_data   |> filter(condition == control_val) |>
      select(value = all_of(out)) |>
      mutate(source = "LLM clone", outcome = outcome_label_map[out])
  )
}) |>
  filter(!is.na(value)) |>
  ggplot(aes(x = value, fill = source, color = source)) +
  geom_density(alpha = 0.35, linewidth = 0.4) +
  scale_fill_manual(values  = source_colors, name = NULL) +
  scale_color_manual(values = source_colors, name = NULL) +
```

```r
  facet_wrap(~ outcome, scales = "free", ncol = 3) +
  labs(x = "Response", y = "Density") +
  plot_theme +
  theme(legend.position = "top")
```

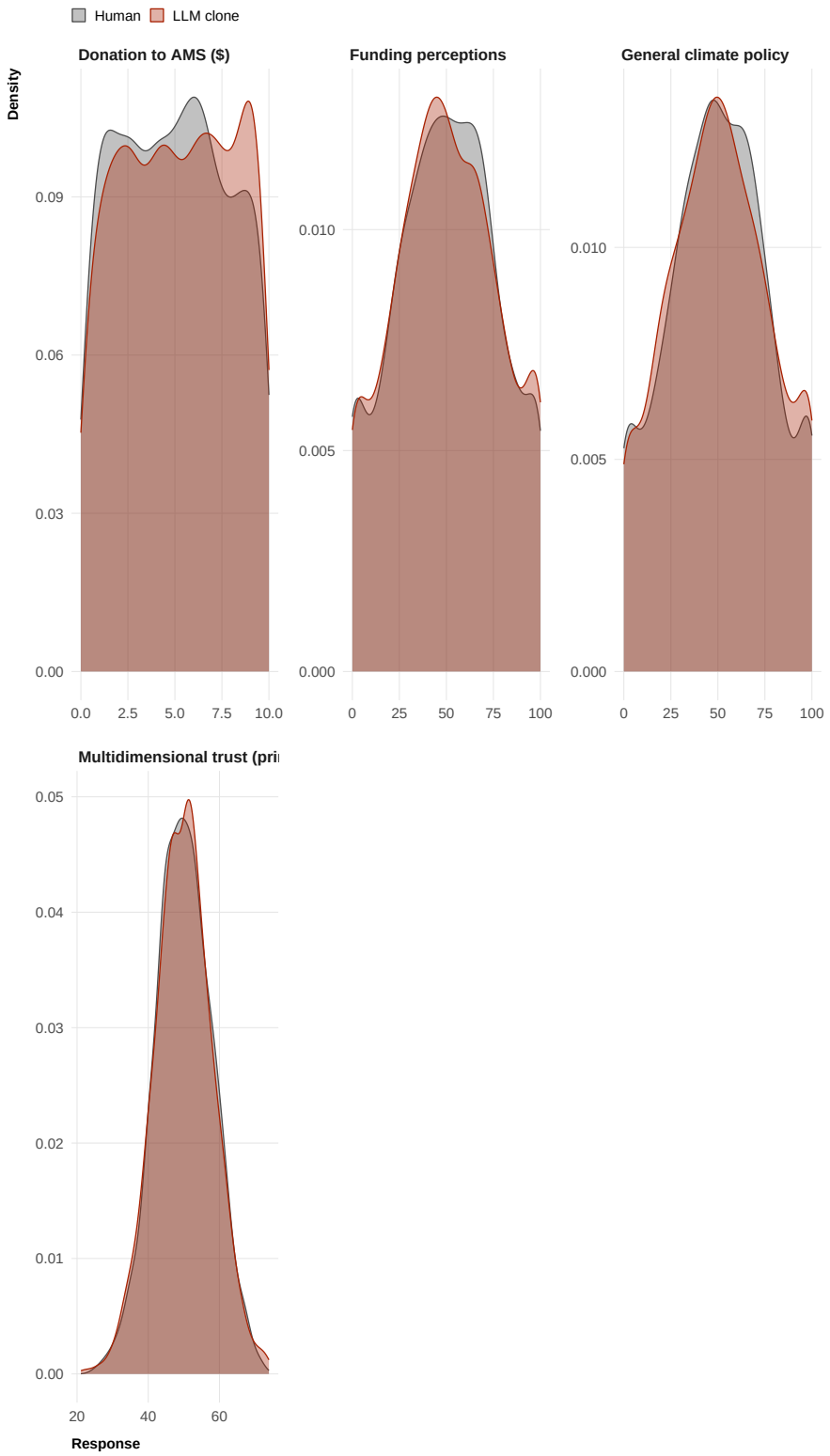

Figure 4: Response distributions in the control condition, by outcome. Human participants shown in dark grey, LLM clones in orange. Each panel covers the full range of that outcome's scale.

```r
dist_control |>
  arrange(match(tier, c("Primary", "Secondary", "Tertiary")), outcome_label) |>
  select(Tier = tier, Outcome = outcome_label,
         OVL = ovl, `KS D` = ks_d, `Variance ratio` = variance_ratio) |>
  gt(groupname_col = "Tier") |>
  fmt_number(columns = c(OVL, `KS D`, `Variance ratio`), decimals = 3) |>
  tab_style(
    style     = cell_text(weight = "bold"),
    locations = cells_row_groups()
  )
```

Table 5: Response distribution metrics in the control condition, by outcome. OVL: overlapping coefficient (1 = identical). KS D: Kolmogorov–Smirnov D-statistic (0 = identical CDFs). Variance ratio: clone / human variance (1 = equal spread).

Outcome	OVL	KS D	Variance ratio
Primary			
Multidimensional trust (primary)	0.974	0.016	1.051
Secondary			
Funding perceptions	0.913	0.019	1.019
Tertiary			
Donation to AMS ($)	0.918	0.042	1.024
General climate policy	0.911	0.019	1.026

2. Subgroup Effects

Heterogeneity in Treatment Effects

Interventions may work differently for different demographic groups — for example, a message about scientific consensus might shift trust more strongly among Republicans than Democrats, or more among younger than older respondents. We ask whether clones reproduce these subgroup differences, or whether they produce a one-size-fits-all response that averages across demographic groups.

For each of six categorical moderators (gender, age band, race, education, income, and partisan identity), we estimate condition × moderator interactions using `run_moderator_model()` separately on human and clone data for all continuous outcomes. Each interaction term captures how much more (or less) effective a given intervention is for one demographic

group compared to the reference group. We compare the full set of interaction estimates between humans and clones using `compare_estimates()` (with `join_by = c("condition", "moderator_level")`). We first report metrics pooled across all illustrative outcomes and moderators, then break down by outcome.

```
# Nested: outcome → moderator → model (illustrative subset)
mod_results_all_h <- map(set_names(outcomes_illustrative), function(out) {
  map(set_names(moderators_cat), function(mod) {
    run_moderator_model(human_data, outcome = out, moderator = mod)
  })
})
mod_results_all_l <- map(set_names(outcomes_illustrative), function(out) {
  map(set_names(moderators_cat), function(mod) {
    run_moderator_model(llm_data, outcome = out, moderator = mod)
  })
})

# Primary-outcome references for the scatter plot below
mod_results_h <- mod_results_all_h[["trust_multidimensional"]]
mod_results_l <- mod_results_all_l[["trust_multidimensional"]]
```

```
# All interaction estimate pairs (outcome × moderator × condition × level)
rq3_pooled_interactions <- map_dfr(outcomes_illustrative, function(out) {
  map_dfr(moderators_cat, function(mod) {
    inner_join(
      mod_results_all_h[[out]][[mod]]$interaction_effects |>
        select(condition, moderator_level, estimate_h = estimate),
      mod_results_all_l[[out]][[mod]]$interaction_effects |>
        select(condition, moderator_level, estimate_l = estimate),
      by = c("condition", "moderator_level")
    ) |> mutate(outcome = out, moderator = mod)
  })
})

# Pooled headline (scale-invariant metrics only; RMSE omitted across mixed scales)
rq3_pooled_row <- rq3_pooled_interactions |>
  summarise(
    spearman_rho    = cor(estimate_h, estimate_l, method = "spearman",
                          use = "pairwise.complete.obs"),
    pearson_r       = cor(estimate_h, estimate_l, use = "pairwise.complete.obs"),
    directional_pct = mean(sign(estimate_h) == sign(estimate_l), na.rm = TRUE) * 100
  ) |>
```

```r
  mutate(outcome_label = "All outcomes", tier = "All outcomes")

# Per-outcome × per-moderator metrics
rq3_metrics_by_outcome <- map_dfr(outcomes_illustrative, function(out) {
  map_dfr(moderators_cat, function(mod) {
    compare_estimates(
      mod_results_all_h[[out]][[mod]]$interaction_effects,
      mod_results_all_l[[out]][[mod]]$interaction_effects,
      join_by = c("condition", "moderator_level")
    ) |> mutate(
      outcome       = out,
      moderator     = mod,
      outcome_label = outcome_label_map[out],
      tier = case_when(
        out == outcomes_primary     ~ "Primary",
        out %in% outcomes_secondary ~ "Secondary",
        TRUE                        ~ "Tertiary"
      )
    )
  })
})
```

Pooled across all outcomes and moderators, Spearman = -0.081, Pearson r = -0.113, directional agreement = 49.2%.

```r
rq3_metrics_by_outcome |>
  arrange(match(tier, c("Primary", "Secondary", "Tertiary")), outcome_label, moderator) |>
  select(Tier = tier, Outcome = outcome_label, Moderator = moderator,
         `Spearman  ` = spearman_rho, `Pearson r` = pearson_r, RMSE = rmse,
         `Directional (%)` = directional_pct,
         `Inferential (%)` = inferential_pct, `TOST (%)` = tost_pct) |>
  gt(groupname_col = "Tier") |>
  fmt_number(columns = c(`Spearman  `, `Pearson r`, RMSE), decimals = 3) |>
  fmt_number(columns = c(`Directional (%)`, `Inferential (%)`, `TOST (%)`), decimals = 1) |>
  tab_style(
    style     = cell_text(weight = "bold"),
    locations = cells_row_groups()
  )
```

Table 6: Interaction estimate comparison metrics by outcome and moderator. Outcomes are grouped by tier; rows are moderators. Spearman : rank correlation across all condition × moderator level pairs. RMSE is in outcome units.

Outcome	Moderator	Spearman	Pearson r	RMSE	Directional (%)	In
Primary						
Multidimensional trust (primary)	age_band	0.085	0.051	1.449	57.0	
Multidimensional trust (primary)	education	-0.100	-0.077	1.470	40.0	
Multidimensional trust (primary)	gender	0.280	0.290	1.042	62.5	
Multidimensional trust (primary)	income	-0.161	-0.192	1.488	51.2	
Multidimensional trust (primary)	party	-0.067	-0.040	1.175	53.3	
Multidimensional trust (primary)	race	0.091	0.132	1.737	50.0	
Secondary						
Funding perceptions	age_band	-0.308	-0.305	5.844	34.0	
Funding perceptions	education	0.123	0.059	5.593	49.0	
Funding perceptions	gender	-0.149	-0.059	5.658	32.5	
Funding perceptions	income	0.018	0.031	5.004	47.5	
Funding perceptions	party	0.220	0.180	4.537	53.3	
Funding perceptions	race	0.111	0.082	4.977	45.0	
Tertiary						
Donation to AMS ($)	age_band	-0.031	-0.007	0.487	47.0	
Donation to AMS ($)	education	0.081	0.066	0.491	74.0	
Donation to AMS ($)	gender	0.200	0.188	0.308	57.5	
Donation to AMS ($)	income	0.193	0.194	0.589	43.8	
Donation to AMS ($)	party	0.118	0.158	0.608	46.7	
Donation to AMS ($)	race	0.005	0.023	0.470	58.8	
General climate policy	age_band	-0.112	-0.116	6.943	37.0	
General climate policy	education	0.004	0.036	5.632	42.0	
General climate policy	gender	0.071	0.118	2.570	57.5	
General climate policy	income	-0.093	-0.070	3.994	53.8	
General climate policy	party	0.090	0.129	4.468	51.7	
General climate policy	race	0.033	0.023	4.943	47.5	

```
map_dfr(moderators_cat, function(mod) {
  inner_join(
    mod_results_h[[mod]]$interaction_effects |>
      select(condition, moderator_level, estimate_h = estimate),
```

```r
    mod_results_l[[mod]]$interaction_effects |>
      select(condition, moderator_level, estimate_l = estimate),
    by = c("condition", "moderator_level")
  ) |> mutate(moderator = mod)
}) |>
  ggplot(aes(x = estimate_h, y = estimate_l)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed",
              color = "grey60", linewidth = 0.4, alpha = 0.35) +
  geom_hline(yintercept = 0, linetype = "dotted", color = "grey80") +
  geom_vline(xintercept = 0, linetype = "dotted", color = "grey80") +
  geom_smooth(method = "lm", se = FALSE, color = "#2980B9", linewidth = 0.6) +
  geom_point(size = 1.5, alpha = 0.65, color = "grey20") +
  facet_wrap(~ moderator, scales = "free", ncol = 2) +
  labs(x = "Human interaction estimate", y = "LLM clone interaction estimate") +
  plot_theme
```

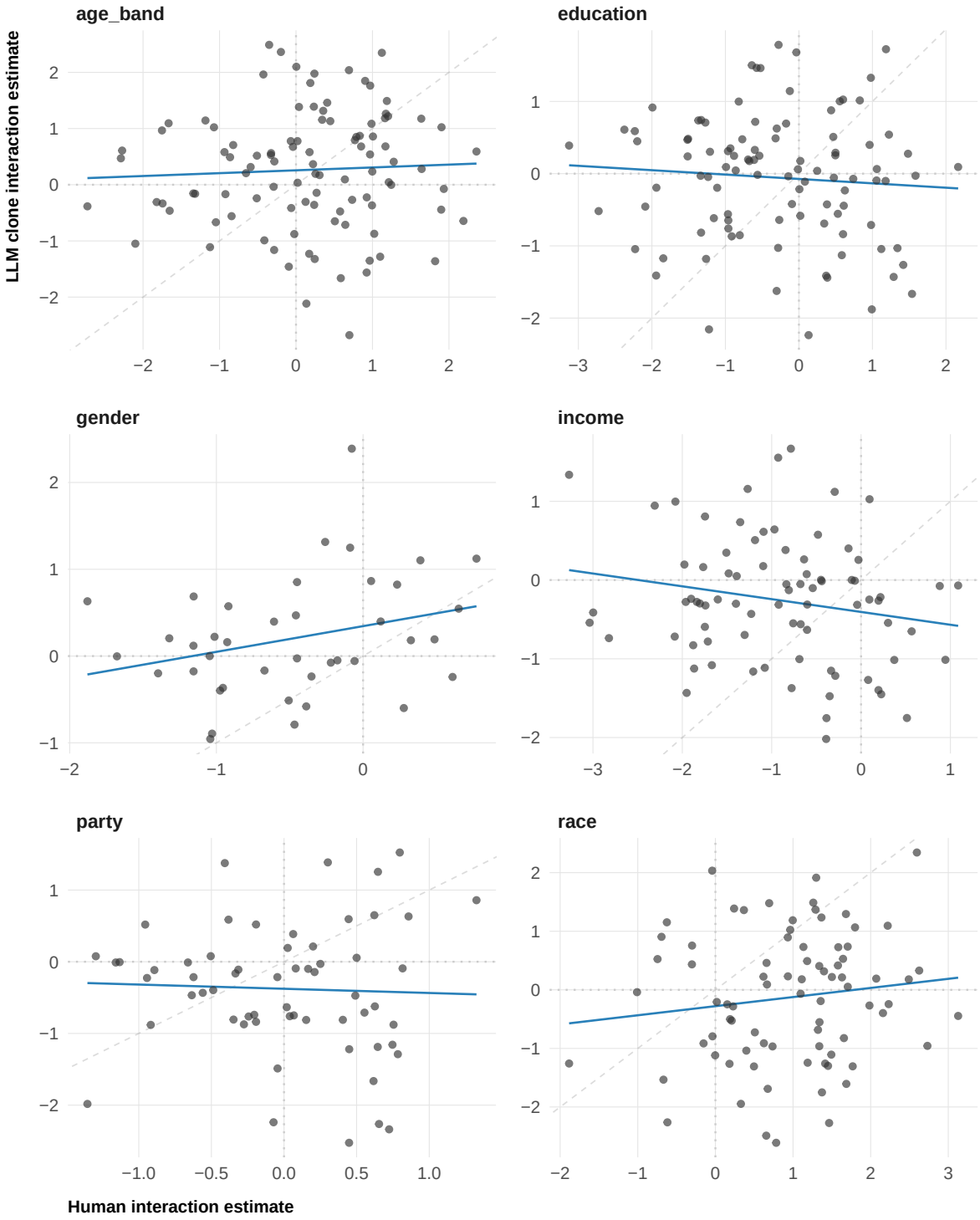

Figure 5: Human vs. LLM clone condition × moderator interaction estimates for the primary outcome (multidimensional trust), by moderator. Each point is one condition × moderator level combination. Blue line: OLS regression. Faint dashed line: perfect calibration (slope = 1).

Response Distributions by Subgroup

Mirroring the distribution analysis in Section 1, we compare response distributions within each demographic subgroup in the control condition for the primary outcome. A clone dataset could reproduce the overall response distribution while having miscalibrated within-group spreads. We use the same OVL, KS D, and variance ratio metrics. Minimum group size of 10 is required for stable kernel density estimation.

```
# For each moderator × level, compare human and clone distributions in control
subgroup_dist <- map_dfr(moderators_cat, function(mod) {
  groups <- levels(human_data[[mod]])
  map_dfr(groups, function(grp) {
    x <- human_data |>
      filter(condition == control_val, .data[[mod]] == grp) |>
      pull(trust_multidimensional) |> na.omit()
    y <- llm_data |>
      filter(condition == control_val, .data[[mod]] == grp) |>
      pull(trust_multidimensional) |> na.omit()
    if (length(x) < 10 | length(y) < 10) return(NULL)
    tibble(moderator = mod, group = grp,
           ovl = compute_ovl(x, y),
           ks_d = suppressWarnings(ks.test(x, y)$statistic),
           variance_ratio = var(y) / var(x))
  })
})
```

```
map_dfr(moderators_cat, function(mod) {
  groups <- levels(human_data[[mod]])
  map_dfr(groups, function(grp) {
    bind_rows(
      human_data |> filter(condition == control_val, .data[[mod]] == grp) |>
        select(value = trust_multidimensional) |>
        mutate(source = "Human",     panel = paste0(mod, ": ", grp)),
      llm_data   |> filter(condition == control_val, .data[[mod]] == grp) |>
        select(value = trust_multidimensional) |>
        mutate(source = "LLM clone", panel = paste0(mod, ": ", grp))
    )
  })
}) |>
  filter(!is.na(value)) |>
  ggplot(aes(x = value, fill = source, color = source)) +
  geom_density(alpha = 0.35, linewidth = 0.4) +
```

```r
  scale_fill_manual(values  = source_colors, name = NULL) +
  scale_color_manual(values = source_colors, name = NULL) +
  facet_wrap(~ panel, scales = "free_y", ncol = 4) +
  labs(x = "Multidimensional trust (0-100)", y = "Density") +
  plot_theme +
  theme(legend.position = "top")
```

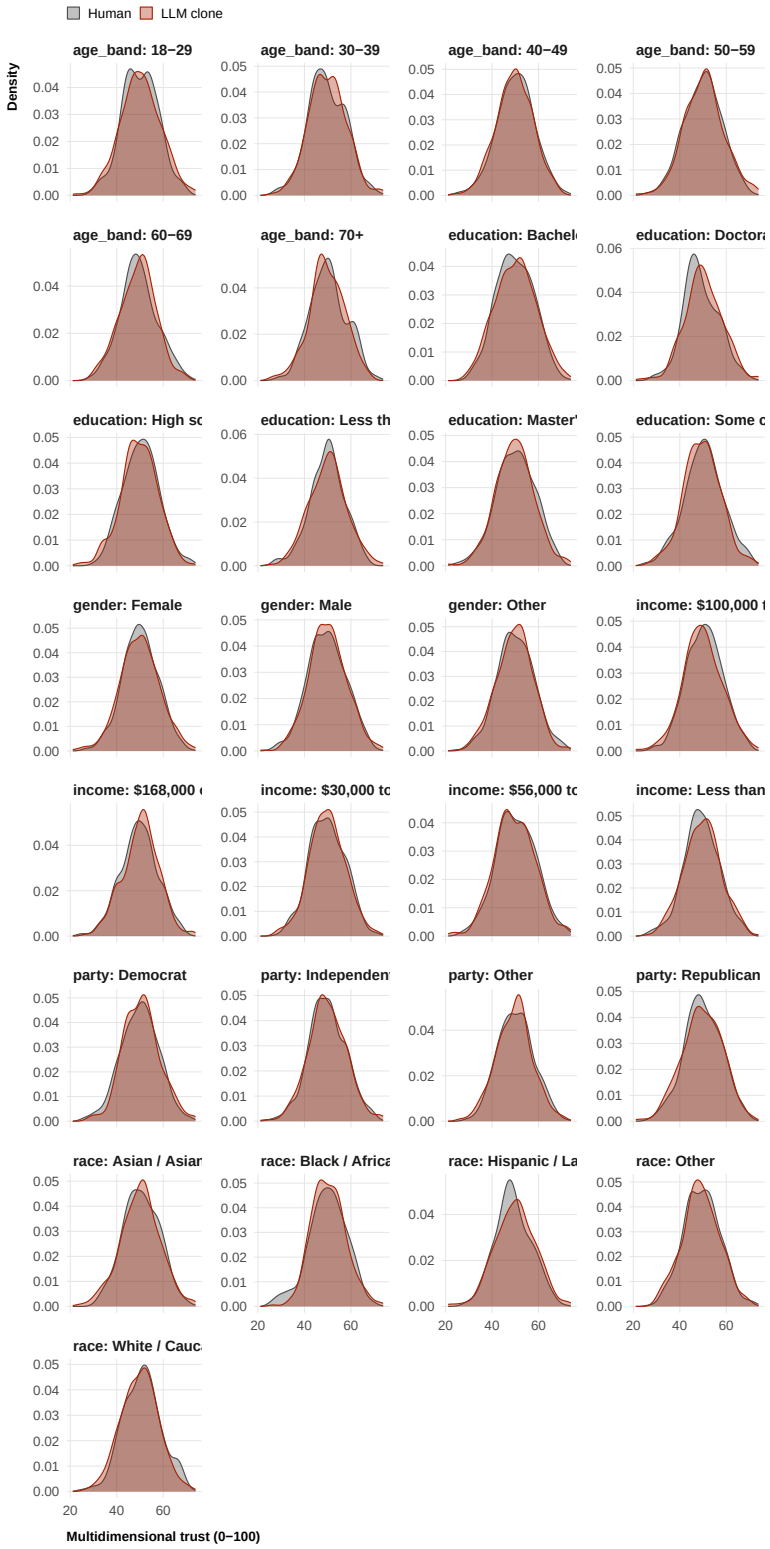

Figure 6: Response distributions within demographic subgroups in the control condition (primary outcome: multidimensional trust). Human participants in dark grey, LLM clones in orange.

```r
subgroup_dist |>
  arrange(moderator, group) |>
  gt(groupname_col = "moderator") |>
  cols_label(group = "Group", ovl = "OVL", ks_d = "KS D",
             variance_ratio = "Variance ratio") |>
  fmt_number(columns = c(ovl, ks_d, variance_ratio), decimals = 3) |>
  tab_style(style = cell_text(weight = "bold"),
            locations = cells_row_groups())
```

Table 7: Within-subgroup distribution comparison in the control condition. OVL: overlapping coefficient (1 = identical). KS D: Kolmogorov–Smirnov statistic. Variance ratio: clone / human.

Group	OVL	KS D	Variance ratio
age_band			
18-29	0.927	0.051	1.253
30-39	0.945	0.063	0.999
40-49	0.963	0.051	0.967
50-59	0.961	0.050	1.130
60-69	0.915	0.068	0.940
70+	0.916	0.071	0.951
education			
Bachelor's degree	0.939	0.056	1.188
Doctorate degree / Ph.D.	0.897	0.117	1.141
High school diploma / GED	0.941	0.067	1.061
Less than high school	0.934	0.046	1.157
Master's degree / Professional degree	0.943	0.060	0.960
Some college or Associate's degree	0.939	0.078	0.885
gender			
Female	0.951	0.050	1.226
Male	0.954	0.049	0.976
Other	0.949	0.043	0.964
income			
$100,000 to $167,999	0.939	0.081	1.183
$168,000 or more	0.933	0.075	0.969
$30,000 to $55,999	0.953	0.048	1.030
$56,000 to $99,999	0.962	0.042	1.041
Less than $30,000	0.939	0.057	1.096
party			
Democrat	0.939	0.058	0.949
Independent	0.963	0.044	1.046
Other	0.944	0.054	1.043
Republican	0.950	0.052	1.153
race			
Asian / Asian American	0.941	0.053	1.209
Black / African American	0.924	0.048	0.811
Hispanic / Latino	0.916	0.107	1.252
Other	0.950	0.044	1.047
White / Caucasian	0.949	0.071	0.968

3. Demographic Stereotyping

Sections 1 and 2 operate on aggregated estimates — average treatment effects, interaction effects, and marginal distributions. Section 3 asks a different question: are individual clone responses calibrated to the demographic profile they were assigned? A clone given a profile of, say, a young Black woman with low income should give responses that match what real young Black women with low income give — not just what the average American gives.

All analyses use the primary outcome (`trust_multidimensional`) and the control condition.

Demographic Baseline Calibration

For each moderator, we compute mean outcomes within each demographic group in the control condition separately for humans and clones, then correlate and compare those cell means. High correlation and low RMSE indicate that clones start from the right baseline response level for each demographic type — before any intervention is applied.

```
# Cell-mean comparison in the control condition, one moderator at a time
baseline_cal <- compare_demographic_baselines(
  human_data, llm_data,
  outcome    = "trust_multidimensional",
  moderators = moderators_cat
)
```

```
baseline_cal |>
  select(Moderator = moderator, r, RMSE = rmse, `N cells` = n_cells) |>
  gt() |>
  fmt_number(columns = c(r, RMSE), decimals = 3)
```

Table 8: Demographic baseline calibration. For each moderator, cell means in the control condition are compared between humans and clones. r: Pearson correlation of cell means. RMSE: root mean squared error. N cells: number of demographic groups.

Moderator	r	RMSE	N cells
gender	-0.797	0.357	3
age_band	0.468	0.436	6
race	0.302	0.681	5
education	-0.966	0.684	6
income	-0.143	0.530	5
party	-0.731	0.511	4

Demographic Predictability

We regress the primary outcome on all six demographic moderators plus condition fixed effects separately for humans and clones. If clones over-rely on demographic stereotypes — mapping demographic variables to outcomes too mechanically — they will show a much higher R^2 than humans: demographics explain more of the variance in clone responses than in human responses. We also compare individual regression coefficients to identify which demographic predictors are over- or underweighted by the clones.

```
pred_test <- compare_demographic_predictability(
  human_data, llm_data,
  outcome    = "trust_multidimensional",
  predictors = moderators_cat
)
```

```
pred_test$r_squared |>
  mutate(source = if_else(source == "human", "Human", "LLM clone")) |>
  gt() |>
  cols_label(source = "Source", r_squared = "R²") |>
  fmt_number(columns = r_squared, decimals = 3)
```

Table 9: R^2 from regressing the primary outcome on demographics + condition FE, separately for humans and clones. Clone R^2 » human R^2 signals demographic stereotyping.

Source	R^2
Human	0.002
LLM clone	0.002

```
pred_test$coefficients |>
  ggplot(aes(x = est_h, y = est_l)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed",
              color = "grey60", linewidth = 0.5) +
  geom_hline(yintercept = 0, linetype = "dotted", color = "grey80") +
  geom_vline(xintercept = 0, linetype = "dotted", color = "grey80") +
  geom_point(size = 2, alpha = 0.8, color = "grey20") +
  geom_text(aes(label = term), size = 2.2, vjust = -0.8, color = "grey40") +
  labs(x = "Human coefficient", y = "LLM clone coefficient") +
  plot_theme
```

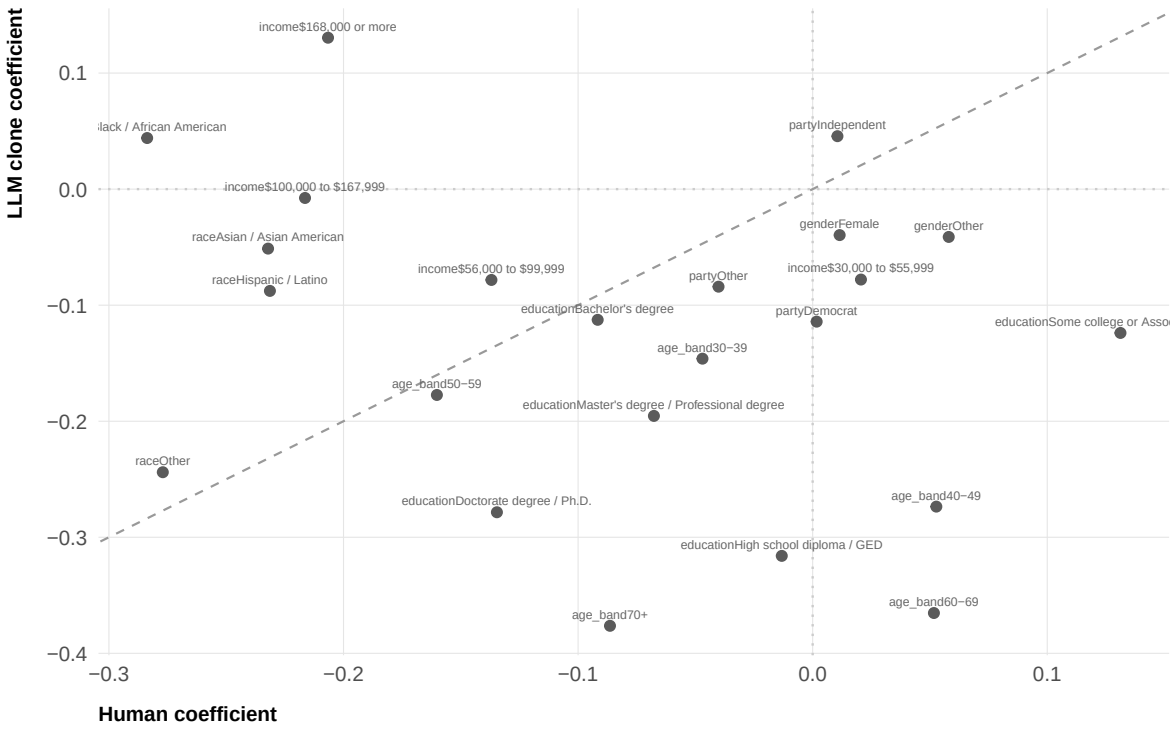

Figure 7: Demographic regression coefficients for the primary outcome in humans (x-axis) vs. LLM clones (y-axis). Each point is one dummy-coded level of a moderator. The dashed line marks perfect agreement.

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